NCC-ZER0 RUNTIME

Test Verification Report

The Neuro-Catalyst Engine | 2026-03-15 | Independent Verification

Aggregate Summary

Total Tests Run	690
Tests Passed	690
Tests Failed	0
Tests Skipped	0
Pass Rate	100.00%
Total Wall Time	6.510s (full suite)
Hardware	Intel Core i5 Gen 8th
GPU	None
Build Profile	Release [optimized]
Test Runner	cargo-nextest
Language	Rust
Date	2026-03-15

Per-Suite Breakdown

Suite	Tests	Duration	Status
Full Integration Suite	**690 / 20 binaries**	6.510s	ALL PASS
Signal Layer	**158 / 7 binaries**	0.161s	ALL PASS
Reputation	**56 / 3 binaries**	5.061s	ALL PASS
TM Layer	**30 / 2 binaries**	0.080s	ALL PASS
AI Provider Adapters	**65 / 2 binaries**	0.059s	ALL PASS
Pipeline	**49 / 3 binaries**	0.920s	ALL PASS
Configurations	**23 / 1 binary**	0.303s	ALL PASS

Notable Test Highlights

The following tests demonstrate governance depth beyond standard unit testing:

Feature #1

Long-running simulation test (5.033s) validating that reputation decay and deterrence mechanisms hold under sustained adversarial pressure. Not a unit test — a behavioral proof.

Feature #2

Extended governance simulation (0.102s) confirming that veto logic, charter axioms, and tribunal procedures remain coherent across a full session lifecycle.

Feature #3

Confirms that the governance layer intercepts and blocks prohibited tool patterns at the kernel level before any execution occurs.

Feature #4

Confirms pre-token short-circuit behavior: deterministic logic units resolve without reaching the language model, demonstrating the pre-cognitive architecture in operation.

Architecture Note

These results were produced on commodity hardware with no GPU acceleration. The test suite covers 20 binaries spanning signal ingestion, intent classification, governance enforcement, memory persistence, provider adapters, and runtime pipeline behavior.

Performance characteristics reflect architecture efficiency, not hardware advantage. The same results are expected to reproduce on equivalent or lesser hardware.